$$\int \frac{1}{\sqrt{x^2+x}}\,dx$$

Integration of Rational Functions by Partial Fractions (7.4)

A **rational function** is a ratio of polynomials:

$$\frac{P(x)}{Q(x)}, \quad \text{where } P \text{ and } Q \text{ are polynomials.}$$

The method of **partial fractions** gives a way to evaluate the integral of any rational function.

Fact: Any rational function $P(x)/Q(x)$ can be written as

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}, \quad \deg R < \deg Q$$

using polynomial division.

$$\int \frac{x^2+x+5}{x+2}\,dx$$

Fact: Any real polynomial can be factored into linear and irreducible quadratic factors:

$$P(x) = c(x-r_1)(x-r_2)\cdots(x^2+a_1x+b_1)(x^2+a_2x+b_2)\cdots$$

Fact: If P, Q are real polynomials and

$$Q(x) = (x-r_1)^{e_1}(x-r_2)^{e_2}\cdots(x^2+a_1x+b_1)^{f_1}(x^2+a_2x+b_2)^{f_2}\cdots$$

then there are constants $A, B, C, \ldots$ such that

$$\frac{P(x)}{Q(x)} = \frac{A}{x-r_1} + \frac{B}{(x-r_1)^2} + \cdots + \frac{C}{(x-r_1)^{e_1}}$$

$$+ \frac{D}{x-r_2} + \frac{E}{(x-r_2)^2} + \cdots + \frac{F}{(x-r_2)^{e_2}}$$

$$\vdots$$

$$+ \frac{Gx+H}{x^2+a_1x+b_1} + \frac{Ix+J}{(x^2+a_1x+b_1)^2} + \cdots + \frac{Kx+L}{(x^2+a_1x+b_1)^{f_1}} + \cdots$$

Example:

$$\frac{x}{(x^4-1)^2}$$

$$\frac{x}{(x^4-1)^2} = \frac{-1/8}{x-1} + \frac{1/16}{(x-1)^2} + \frac{-1/8}{x+1} + \frac{-1/16}{(x+1)^2}$$

$$+ \frac{(1/4)x}{x^2+1} + \frac{(1/4)x}{(x^2+1)^2}$$

$$\int \frac{x-12}{x^2-4x}\,dx$$

$$\int \frac{5x+1}{2x^2-x-1}\,dx$$

$$\int \frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$$

$$\int \frac{3x^2 - x + 8}{x^3 + 4x} \, dx$$

Fact you can use:

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

$$\int \frac{x+4}{x^2+2x+6}\,dx$$

$$\int \frac{1}{2\sqrt{x+3}+x}\,dx$$